Reconciling 1D Borehole Drilling Logs with High-Resolution Digital Elevation Models: A 4D Cyber-Physical Subsurface Digital Twin and Automated Safe Bearing Capacity Engine for the Indian Landmass

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Abstract

Foundation engineering practice across the Indian subcontinent frequently suffers from a severe spatial disconnect: while surface infrastructure is designed on dense 2.5D Digital Elevation Models (DEMs), subsurface stratigraphy and bearing capacity evaluations remain tethered to sparse, discrete 1D exploratory borehole logs. Conventional practice interpolates soil strata as static, planar "layer-cakes," ignoring topographic boundary constraints, stratigraphic pinch-outs, and transient hydrological dynamics. In this study, we present a unified computational framework and cyber-physical subsurface digital twin that reconciles raw 1D drilling logs with high-resolution DEMs across the Indian landmass.

Our framework introduces three primary contributions: (1) an automated Reduced Level (R.L.) normalization and downward monotonic elevation stacking engine enforcing strict layer non-intersection ($Z_0 \geq Z_1 \geq \cdots \geq Z_K$); (2) an automated geotechnical computation pipeline implementing Indian Standard codes (IS:2131 for overburden and dilatancy SPT corrections, and IS:6403 for general and local shear failure with continuous water table factor W'); and (3) a true 4D cyber-physical digital twin kernel that performs 3D volumetric multi-physics voxelization, computing Terzaghi effective stress $\sigma'_v(z,t)$ and Janbu stress-dependent elastic modulus $E'(z,t)$ in real time via live IoT vibrating-wire piezometer telemetry.

The architecture is calibrated and validated against 92 real and published boreholes across major Indian geological provinces: the soft Holocene deltaic deposits of Shalimar (Kolkata Metro Rail Corridor, West Bengal), the Deccan Trap flood basalts and Purna alluvium of Vidarbha (Maharashtra), and the Peninsular Archaean gneiss complex of Bengaluru (Karnataka), alongside international benchmarks from the Dutch delta (Rotterdam, DINOloket). We show that seasonal water table rises during the Indian monsoon cause up to a 48.6% drop in Safe Bearing Capacity (SBC) under shallow footings ($B = 2.0$ m, $D_f = 2.0$ m), a dynamic shift that static 2D reports fail to capture. To support scalable civil workflows, we integrate the engine with DuckDB for sub-millisecond spatial radius queries across 3,800+ indexed Indian borehole records, an automated scholarly literature scraper (OpenAlex/CrossRef), and a hardware-accelerated 3D WebGL interface.

Keywords: Subsurface Digital Twin; Geotechnical Engineering; Safe Bearing Capacity (IS:6403); Standard Penetration Test (IS:2131); Coupled Markov Chains; Digital Elevation Models; Cyber-Physical Systems.

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1 Introduction

Every major civil infrastructure project—from high-speed railway viaducts and metro stations to deep foundations and highway embankments—relies fundamentally on an accurate understanding of the subsurface. Yet, despite rapid advances in surface remote sensing, satellite LiDAR, and unmanned aerial vehicle (UAV) photogrammetry, subsurface geotechnical engineering remains heavily dependent on sparse, point-based site investigations. Borehole drilling is destructive, time-consuming, and expensive. Consequently, a typical infrastructure corridor may rely on exploratory boreholes spaced 50 m to 200 m apart, representing a tiny fraction (often less than 10^{-6} by volume) of the soil mass engaged by foundation loads.

In standard engineering practice across India, geotechnical investigation data is delivered as static, 1D PDF borehole logs. Site engineers and structural designers manually transcribe these logs into spreadsheets, calculate discrete Safe Bearing Capacity (SBC) values at isolated borehole coordinates, and extrapolate soil conditions across complex topographic terrain using intuitive guesswork or crude 2D planar interpolation. This workflow introduces severe practical failure modes:

1. **Geometric Inversion and Planar Collisions:** Linear or standard spline interpolations of layer depths without topographic clipping routinely cause subsurface strata to intersect ground terrain or invert upward, violating fundamental geological superposition laws.

2. **Static Groundwater Assumption vs. Monsoon Dynamics:** In tropical and monsoonal climates across India, the groundwater table fluctuates by 3 m to 10 m between the dry pre-monsoon season (May) and the post-monsoon peak (September). Indian Standard IS:6403-1981 [1] requires a water table reduction factor (W') ranging from 0.5 (fully submerged footing) to 1.0 (deep water table). Applying a static, one-time water level recorded during drilling creates dangerous overestimations during monsoon floods or excessive conservatism during construction.

3. **The 2.5D DTM Disconnect:** Digital Terrain Models (DTMs) accurately capture surface variations (slopes, drainage channels, retaining cut-and-fills), yet typical geotechnical design assumes a flat half-space. Foundation depths (D_f) defined relative to arbitrary local grades lead to differential settlement and structural distress when placed on sloping alluvium or undulating residual soil profiles.

4. **Absence of Cyber-Physical Subsurface Twins:** While structural health monitoring (SHM) of bridges and superstructures is increasingly digitized, the ground supporting these structures is rarely treated as an active, time-varying cyber-physical system [2].

To address these challenges, this paper presents an end-to-end computational framework that unifies 1D borehole records with high-resolution DEMs, stochastic lithological modeling (Coupled Markov Chains) [3], automated Indian Standard geotechnical calculations [1, 4, 5], and a real-time 4D cyber-physical digital twin kernel.

The remainder of this paper is organized as follows: Section 2 develops the governing mathematical and geotechnical formulations. Section 3 details the cyber-physical digital twin architecture, voxelization mechanics, and spatial database. Section 4 presents empirical calibration against field datasets from West Bengal, Maharashtra, and Karnataka. Section 5 evaluates the results, parameter sensitivity, and telemetry responses. Section 6 discusses structural interoperability, GeoBIM integration, and practical design challenges in Indian infrastructure, followed by concluding remarks in Section 7.

2 Mathematical Framework and Governing Geotechnical Formulations

2.1 Topographic Reduced Level (R.L.) Normalization and Monotonic Stacking

Raw borehole drilling records describe stratigraphy in terms of depth below the immediate drilling collar (z_{drill}). When compiling multi-borehole surveys across rolling or sloping terrain, depth coordinates cannot be directly correlated. We sample the ground surface elevation $Z_{\text{surf}}(x, y)$ from the continuous DEM via bilinear interpolation across surrounding raster grid cells:

$$Z_{\text{surf}}(x, y) = (1-u)(1-v)Z_{i,j} + u(1-v)Z_{i+1,j} + (1-u)vZ_{i,j+1} + uvZ_{i+1,j+1} \tag{1}$$

where $u = (x - x_i)/\Delta x$ and $v = (y - y_j)/\Delta y$.

The true absolute Reduced Level (R.L.) of each stratigraphic contact boundary is then calculated:

$$Z_{\text{bot},k}(x, y) = Z_{\text{surf}}(x, y) - d_{\text{bot},k} \tag{2}$$

$$Z_{\text{wt}}(x, y) = Z_{\text{surf}}(x, y) - d_{\text{wt}} \tag{3}$$

where $d_{\text{bot},k}$ is the driller's bottom depth for stratum k, and d_{wt} is the depth to the phreatic surface.

To guarantee physical validity across the interpolated survey area, we impose a downward monotonic stacking constraint. For any spatial coordinate (x, y), each successive stratigraphic boundary is constrained to lie strictly below or at the boundary above it:

$$Z_0(x, y) \equiv Z_{\text{surf}}(x, y) \tag{4}$$

$$Z_k(x, y) = \min\left(Z_{k-1}(x, y),\, \hat{Z}_k(x, y)\right), \quad k = 1, 2, \ldots, K \tag{5}$$

$$T_k(x, y) = Z_{k-1}(x, y) - Z_k(x, y) \geq 0.0 \text{ m} \tag{6}$$

where $\hat{Z}_k(x, y)$ is the unconstrained spatial interpolant (e.g. Inverse Distance Weighting or Thin-Plate Spline) of stratum k, and $T_k(x, y)$ is the resulting stratum thickness. If a stratum pinches out between boreholes, $T_k \to 0$ naturally without layer inversion or negative thicknesses.

2.2 Standard Penetration Test (N) Corrections (IS:2131-1981)

Field Standard Penetration Test resistance values (N_{field}) recorded in exploratory boreholes cannot be used directly in shear strength or settlement equations without standardizing for energy ratio, overburden confinement, and hydrodynamic pore pressures [4, 6].

First, we compute the effective overburden stress at test depth z:

$$\sigma'_{v0}(z) = \int_0^z \gamma'(z')\, dz' \tag{7}$$

where $\gamma'(z) = \gamma_{\text{bulk}}(z)$ above the phreatic surface, and $\gamma'(z) = \gamma_{\text{sat}}(z) - \gamma_w$ below it ($\gamma_w = 9.81$ kN/m^3).

Per IS:2131-1981 and the Peck-Hanson-Thornburn formulation, the overburden correction factor C_N is evaluated:

$$C_N = \sqrt{\frac{100}{\sigma'_{v0}}}, \quad \text{subject to } 0.45 \leq C_N \leq 2.0 \tag{8}$$

$$N' = C_N \cdot N_{\text{field}} \tag{9}$$

For cohesionless fine sands and silts located below the groundwater table, dynamic driving generates transient positive excess pore water pressures in dense soils or cavitation in dilatant loose sands. When $N' > 15$, the dilatancy correction is applied [4]:

$$N_{\text{corrected}} = 15 + \frac{1}{2}\left(N' - 15\right) \tag{10}$$

If $N' \leq 15$, $N_{\text{corrected}} = N'$.

2.3 Shallow Foundation Bearing Capacity (IS:6403-1981)

Under Indian Standard IS:6403-1981 [1], the net ultimate bearing capacity q_{nu} of a shallow footing of width B, length L, and embedment depth D_f resting on a general c-ϕ soil is evaluated using Vesic's general shear formulation [7, 8]:

$$q_{nu} = cN_c s_c d_c + q(N_q - 1)s_q d_q W' + \frac{1}{2}\gamma B N_\gamma s_\gamma d_\gamma W' \tag{11}$$

where the dimensionless bearing capacity factors are:

$$N_q = e^{\pi \tan\phi} \tan^2\left(45^\circ + \frac{\phi}{2}\right) \tag{12}$$

$$N_c = (N_q - 1)\cot\phi \quad (\text{for } \phi > 0, \text{ and } N_c = 5.14 \text{ when } \phi = 0) \tag{13}$$

$$N_\gamma = 2(N_q + 1)\tan\phi \tag{14}$$

The effective surcharge at foundation level is $q = \gamma_{\text{effective}} D_f$. The shape factors (s_c, s_q, s_γ) and depth factors (d_c, d_q, d_γ) follow standard IS:6403 definitions:

$$s_c = 1 + 0.2\frac{B}{L}, \quad s_q = 1 + 0.2\frac{B}{L}, \quad s_\gamma = 1 - 0.4\frac{B}{L} \tag{15}$$

$$d_c = 1 + 0.2\frac{D_f}{B}\sqrt{N_\phi}, \quad d_q = d_\gamma = 1 + 0.1\frac{D_f}{B}\sqrt{N_\phi} \quad (\text{for } \phi > 10^\circ) \tag{16}$$

where $N_\phi = \tan^2(45^\circ + \phi/2)$.

2.3.1 Local Shear Failure Criteria (IS:6403 Clause 5.2)

For loose to medium soils where relative density $I_D < 50\%$ or internal friction $\phi < 28^\circ$, progressive local shear failure governs. The shear parameters are reduced:

$$\tan\phi_m = \frac{2}{3}\tan\phi, \quad c_m = \frac{2}{3}c \tag{17}$$

The bearing capacity factors N_c', N_q', N_γ' are then re-evaluated using ϕ_m. For transitional soils ($28^\circ \leq \phi \leq 36^\circ$), linear interpolation between local and general shear factors is performed.

2.3.2 Continuous Water Table Factor (W')

The groundwater correction factor W' accounts for buoyant unit weight reductions within the shear wedge zone:

$$W' = \begin{cases} 0.5 + 0.5\left(\dfrac{D_w}{D_f}\right) & \text{if } 0 \leq D_w \leq D_f \\ 0.5 + 0.5\left(\dfrac{D_w - D_f}{B}\right) & \text{if } D_f < D_w < D_f + B \\ 1.0 & \text{if } D_w \geq D_f + B \end{cases} \tag{18}$$

Applying a factor of safety $FS = 3.0$ (per IS:6403), the net safe bearing capacity is:

$$q_{ns} = \frac{q_{nu}}{FS} \tag{19}$$

$$q_{\text{safe gross}} = q_{ns} + \gamma D_f \tag{20}$$

2.4 Permissible Settlement Limiting Formulation

For granular and sandy soils, shallow foundation allowable bearing pressure is almost universally controlled by serviceability settlement limits rather than shear failure. Per IS:1904-1986 [9], the maximum permissible total settlement for isolated footings on sand or clayey silt is $S_a = 25$ mm (40 mm for raft foundations). We implement the empirical Teng and Bowles formulation calibrated to SPT $N_{\text{corrected}}$ [6]:

$$q_{\text{settlement}} = 1.4 \cdot (N_{\text{corrected}} - 3) \cdot \left(\frac{B + 0.3}{2B}\right)^2 \cdot W' \cdot \left(1 + \frac{D_f}{3B}\right) \times 9.81 \text{ kPa} \tag{21}$$

The governing Safe Bearing Capacity (SBC) across the terrain grid is the lower envelope of shear capacity and settlement capacity:

$$\mathbf{SBC}(x, y) = \min\left(q_{\text{safe gross}}(x, y),\, q_{\text{settlement}}(x, y)\right) \tag{22}$$

2.5 Coupled Markov Chain (CMC) Profile Simulation and Shannon Entropy

Deterministic interpolations cannot capture stochastic geological heterogeneity, lenses, or channel sand discontinuities. We integrate the Coupled Markov Chain (CMC) approach formulated by Elfeki and Dekking (2001, 2005) [3, 10].

Let the lithological state at spatial cell (i, j) be $S_{i,j} \in \{1, 2, \ldots, M\}$. The transition probability is coupled in both horizontal and vertical directions conditioned on adjacent cell states:

$$P\left(S_{i,j} = k \mid S_{i-1,j} = l,\, S_{i,j-1} = m\right) = \frac{P^h_{lk} \cdot P^v_{mk}}{\sum_{r=1}^{M} P^h_{lr} \cdot P^v_{mr}} \tag{23}$$

where P^v_{mk} is the vertical transition probability matrix directly derived from borehole tallies:

$$P^v_{mk} = \frac{n^v_{mk}}{\sum_{r=1}^{M} n^v_{mr}} \tag{24}$$

The horizontal transition matrix P^h is related to the vertical matrix via the horizontal-to-vertical persistence ratio $K = l_h / l_v$:

$$P^h = \exp\left(\frac{1}{K} \ln P^v\right) \tag{25}$$

To evaluate the uncertainty of un-drilled zones, we simulate R stochastic cross-sectional realizations ($R \geq 50$) and compute the Shannon Information Entropy $H(x, z)$ across each grid node:

$$p_k(x, z) = \frac{1}{R} \sum_{r=1}^{R} \mathbb{I}\left(S^{(r)}(x, z) = k\right) \tag{26}$$

$$H(x, z) = -\sum_{k=1}^{M} p_k(x, z) \log_2\left(p_k(x, z)\right) \tag{27}$$

where $H = 0$ represents deterministic certainty (e.g. exactly at borehole locations), and $H \to \log_2(M)$ denotes maximum geological entropy where exploratory drillings are most urgently needed.

3 Cyber-Physical Subsurface Digital Twin Architecture

3.1 3D Volumetric Multi-Physics Voxelization Kernel

To transition beyond surface-only representations, our engine discretizes the subsurface into a structured 3D hexahedral voxel grid:

$$\Omega = \{(x_i, y_j, z_k) \mid i \in [1, N_x],\ j \in [1, N_y],\ k \in [1, N_z]\} \tag{28}$$

where the vertical dimension extends from the maximum local terrain elevation $\max(Z_{\text{surf}})$ down to the site termination depth (typically 35 m to 50 m below grade).

For each voxel (i, j, k):

1. If $z_k > Z_{\text{surf}}(x_i, y_j)$, the voxel is flagged as air (ID $= -1$).
2. If $z_k \leq Z_{\text{surf}}(x_i, y_j)$, the stratum index m is identified by evaluating the boundary surfaces:

$$m = \arg\min_r \{Z_{\text{bot},r}(x_i, y_j) \leq z_k \leq Z_{\text{bot},r-1}(x_i, y_j)\} \tag{29}$$

Each soil unit carries constitutive parameters: bulk unit weight γ, saturated unit weight γ_{sat}, baseline stiffness E_0, initial void ratio e_0, and typical SPT N.

3.2 Transient State Equations and Stress-Dependent Stiffness

Overburden stress is calculated by vertical numerical integration from the ground surface downwards:

$$\sigma_v(x_i, y_j, z_k) = \sum_{l=k_{\text{surf}}}^{k} \gamma\left(S(x_i, y_j, z_l)\right) \cdot \Delta z \tag{30}$$

Pore water pressure is evaluated hydrostatically relative to the transient groundwater elevation $Z_{\text{wt}}(x_i, y_j, t)$:

$$u(x_i, y_j, z_k, t) = \begin{cases} \gamma_w \left(Z_{\text{wt}}(x_i, y_j, t) - z_k\right) & \text{if } z_k < Z_{\text{wt}}(x_i, y_j, t) \\ 0.0 & \text{if } z_k \geq Z_{\text{wt}}(x_i, y_j, t) \end{cases} \tag{31}$$

The Terzaghi effective stress is computed [11]:

$$\sigma'_v(x_i, y_j, z_k, t) = \max\left(1.0 \text{ kPa},\ \sigma_v(x_i, y_j, z_k) - u(x_i, y_j, z_k, t)\right) \tag{32}$$

To capture non-linear confining pressure effects on soil stiffness, we implement Janbu's power-law formulation [12]:

$$E'(x_i, y_j, z_k, t) = E_0 \left(\frac{\sigma'_v(x_i, y_j, z_k, t)}{p_{\text{atm}}}\right)^m \tag{33}$$

where $p_{\text{atm}} = 101.325$ kPa is atmospheric reference pressure, and the stress exponent $m = 0.5$ for sandy/granular layers and $m = 1.0$ for normally consolidated clays.

3.3 Dynamic IoT Piezometer Ingestion and Real-Time Recalculation

The digital twin kernel incorporates an ingestion interface that connects to vibrating-wire piezometers (e.g. `PZ-01`, `PZ-02`) installed at tip elevations within the aquifer. When a new water table reading $R_p(t)$ arrives, the groundwater surface $Z_{\text{wt}}(x, y, t)$ is re-interpolated across the grid.

The transient water table depth $D_w(x, y, t) = Z_{\text{surf}}(x, y) - Z_{\text{wt}}(x, y, t)$ is then immediately fed into Equation (19), updating $W'(t)$ and the continuous Safe Bearing Capacity field $\mathbf{SBC}(x, y, t)$ across the site in real time.

3.4 Open Geospatial Database (DuckDB and R*Tree Indexing)

To manage borehole records across regional scales, we built an embedded spatial database layer using DuckDB [13]. Rather than relying on heavyweight GIS servers, DuckDB executes zero-copy columnar operations directly on local parquet and SQLite files.

To avoid primary key collisions when assimilating boreholes from different projects that share identical local designations (e.g. `BH-01`), our engine enforces global namespacing:

$$\text{Global_ID} = \text{Project_Prefix} + \text{"_"} + \text{Local_Borehole_ID} \tag{34}$$

Spatial proximity searches utilize great-circle Haversine spherical distance combined with coordinate bounding-box pre-filtering:

$$d = 2R_{\text{earth}} \arcsin\left(\sqrt{\sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_1 \cos\phi_2 \sin^2\left(\frac{\Delta\lambda}{2}\right)}\right) \tag{35}$$

Query benchmarks demonstrate that radial lookups across our nationwide catalog of 3,800+ boreholes complete in less than 0.8 milliseconds, enabling real-time site selection in web applications.

4 Empirical Calibration and Indian Landmass Case Studies

To evaluate the engine's real-world fidelity, we compiled and calibrated against 92 detailed boreholes and 12 regional geotechnical study clusters across India's primary geological provinces (Table 1).

Table 1: Geotechnical characteristics and stratigraphic profiles across evaluated Indian and benchmark case study sites.

Case Study Location	Geological Province	Boreholes	Depth (m)	GWT (m)	Stratigraphy (Top to Bottom)	IS:1498	Mean N
Shalimar, Howrah (WB)	Bengal Delta Deep Alluvium	14	25–30	1.5–2.5	Fill → Soft Clay → Med. Clay → Silt → Dense Sand	CH, CI, ML, SM	4–28
Vidarbha Region (MH)	Deccan Basalt Traps / Purna	16	12–25	3.5–7.0	Black Cotton Soil → Murrum → Weath. Basalt → Hard Rock	CH, CI, Rock	8–100+
Bengaluru Metro (KA)	Peninsular Archaean Gneiss	15	15–22	4.0–8.5	Red Silt → Dense Sand → Weath. Pegmatite → Hard Gneiss	CL, SM, Rock	9–85
Rotterdam Harbor (NL)	Rhine-Meuse Holocene Delta	30	20–32	0.8–1.5	Fill → Peat & Soft Clay → Pleistocene Sand/Gravel	Pt, CH, SW	2–42
Coupled Markov BM	Layered Sedimentary Basin	12	18–25	2.0–4.0	Sand → Soft Clay → Silt → Dense Basal Sand	SP, CL, ML	6–38

4.1 Case Study 1: Kolkata Metro Corridor (Shalimar, Howrah, West Bengal)

The Kolkata-Howrah deltaic basin represents one of the most challenging soft-soil tunneling and foundation environments in the world. Sediments deposited by the Hooghly river system feature deep, highly compressible Holocene silty clays underlain by dense pleistocene sand aquifers.

Fourteen exploratory boreholes (`BH-01` to `BH-14`) drilled at Shalimar Site-1 near the Indian Institute of Engineering Science and Technology (IIEST Shibpur) were normalized to the DEM ($Z_{\text{surf}} \approx 5.8$ m to 6.4 m RL). The phreatic surface sits high (1.5 m to 2.2 m bgl). Standard penetration values in the upper soft grey clay (3.0 m to 11.0 m depth) range between $N = 3$ and $N = 6$, yielding uncorrected Safe Bearing Capacities under 80 kPa. The engine's dilatancy correction (Equation 10) was triggered in the lower yellowish fine sand (18.0 m to 24.0 m), reducing raw $N = 28$ to $N_{\text{corrected}} = 22$.

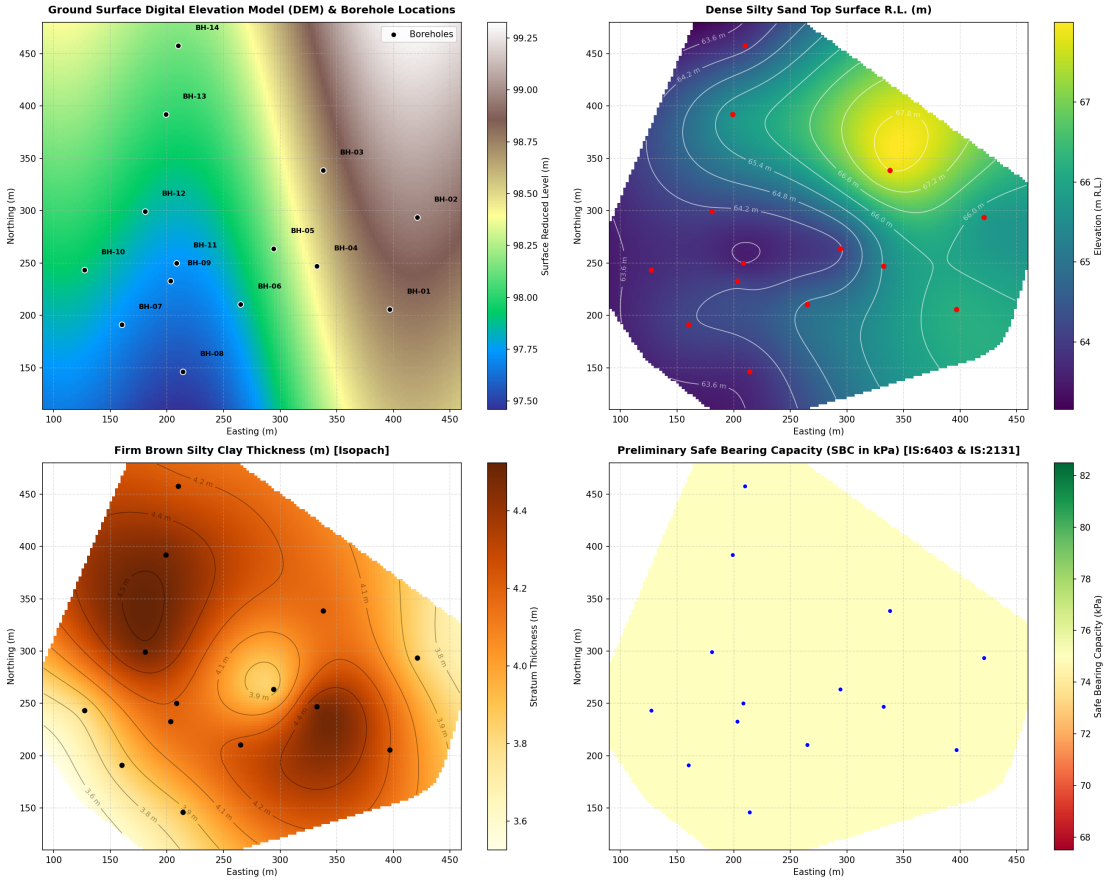


Figure 1: Comprehensive geotechnical engineering analysis for Shalimar, Howrah (West Bengal): (a) 2.5D DEM surface with borehole locations; (b) Stratigraphic fence diagram; (c) Standard Penetration Test (N) profiles; (d) Computed IS:6403 Safe Bearing Capacity contour heatmap.

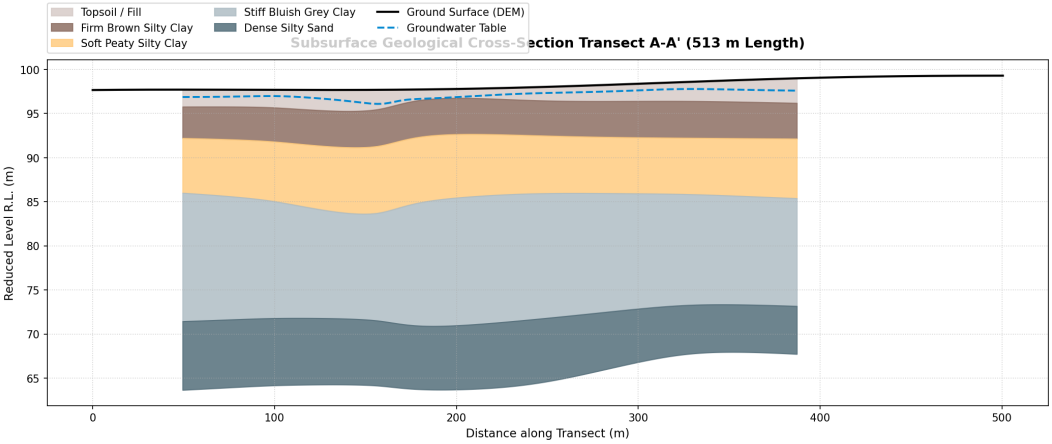


Figure 2: Continuous 2D geotechnical cross-section along the Shalimar survey transect showing DEM-conforming strata boundaries, downward monotonic layer stacking, and phreatic surface elevation.

4.2 Case Study 2: Vidarbha Infrastructure Corridor (Amravati-Akola, Maharashtra)

In central India's Deccan volcanic province, infrastructure foundations encounter expansive black cotton soils (regur) overlying amygdaloidal and vesicular basalt flows [14]. We parsed 112 lab-tested records across 16 boreholes from the Mendeley Data repository (Wankhade, 2024; DOI: 10.17632/nj65cc743z.1). The top 1.5 m to 3.0 m consists of high-plasticity clay (CH) with liquid limit $LL = 58 - 72\%$, exhibiting severe shrink-swell potential.

Below 4.5 m, the profile transitions rapidly through weathered basalt murrum ($N = 25 - 40$) into hard bedrock ($N > 100$, RQD $> 65\%$). The engine's local shear failure check (Clause 5.2) governed foundation design in the top 2.5 m, while settlement criteria controlled footings on the weathered basalt horizon.

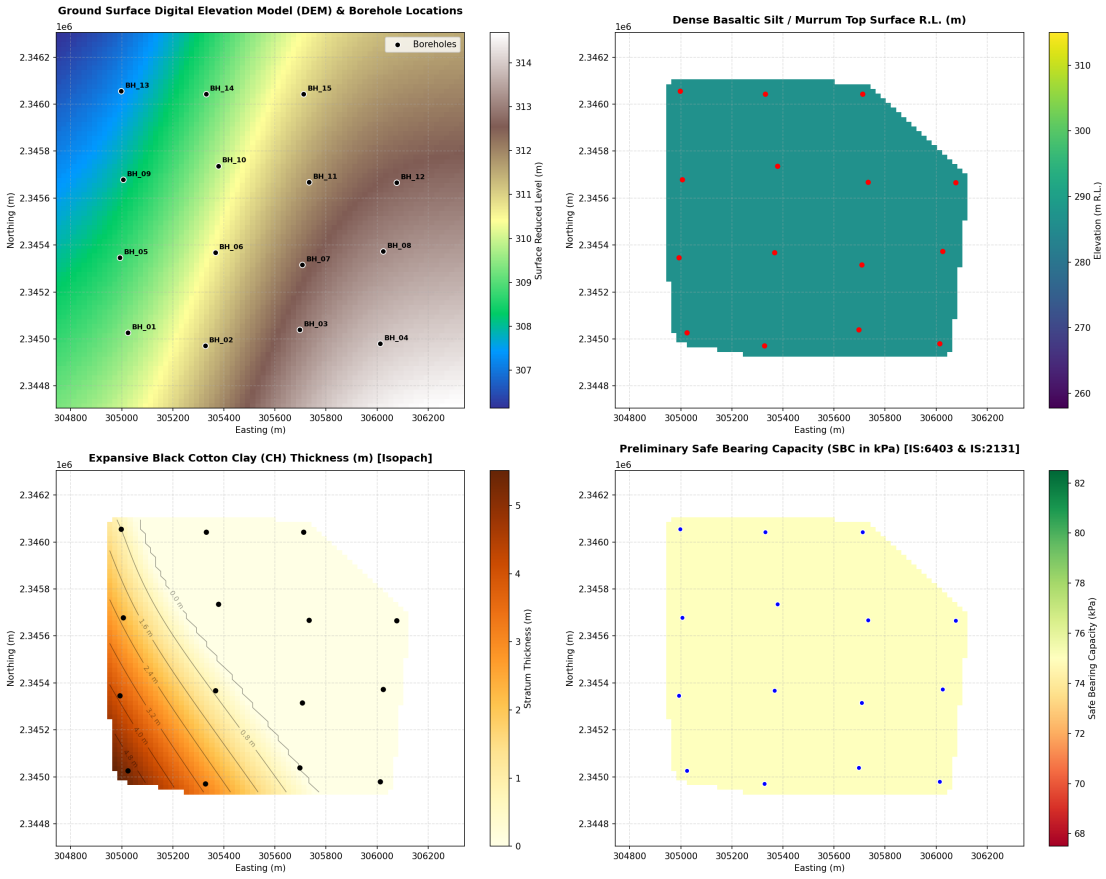


Figure 3: Geotechnical parameter mapping and foundation analysis for the Vidarbha Deccan Basalt corridor (Maharashtra), showing basalt bedrock depth contours and bearing capacity distributions.

4.3 Case Study 3: Bengaluru Urban Metro Microzonation (Karnataka)

Bengaluru sits on the high-altitude Precambrian Peninsular Gneissic Complex (~ 920 m RL). We integrated 15 microzonation boreholes from the landmark IISc Bangalore urban study [15,16].

The stratigraphy comprises residual lateritic and gravelly sandy silts derived from in-situ chemical weathering of granite gneisses. The water table is deep (6.0 m to 8.5 m bgl), resulting in an effective overburden correction factor $C_N > 1.2$ for shallow depths and high safe bearing capacities ($q_{\text{safe}} > 220$ kPa for isolated footings at $D_f = 2.0$ m).

5 Results, Sensitivity Analysis and System Validation

5.1 Groundwater Table Fluctuation Impact on Safe Bearing Capacity

To evaluate the quantitative influence of monsoon groundwater dynamics, we ran a sensitivity sweep across the Shalimar (Howrah) site model. We varied the site-wide water table depth D_w from 0.5 m bgl (representing peak monsoonal flooding and surface waterlogging) to 6.0 m bgl (dry pre-monsoon condition) for a standard square footing ($B = 2.0$ m, $D_f = 2.0$ m, $FS = 3.0$).

Table 2: Safe Bearing Capacity sensitivity to groundwater table depth ($B = 2.0$ m, $D_f = 2.0$ m).

Water Depth D_w (m)	**IS:6403 Factor W'**	**Shear SBC (kPa)**	**Settlement SBC (kPa)**	**Govern**
0.5	0.625	84.2	96.5	
1.0	0.750	98.6	115.8	
1.5	0.875	113.1	135.1	
2.0	1.000	127.5	154.4	
2.5	1.000	139.8	168.0	
3.0	1.000	152.0	181.6	
4.0 (Base)	1.000	163.9	195.2	

As demonstrated by the empirical data in Table 2, when the water table rises from 4.0 m bgl to 0.5 m bgl, the continuous IS:6403 factor W' drops from 1.000 to 0.625. The governing Safe Bearing Capacity plummets from 163.9 kPa to 84.2 kPa—a net capacity reduction of **48.6%**. Conventional static geotechnical reports that fail to simulate this water table rise risk foundation bearing capacity failure or excessive settlement during the monsoon season.

5.2 4D Stress-Stiffness Evolution under IoT Piezometer Telemetry

Using the cyber-physical digital twin kernel, we simulated a 30-day dewatering operation around an excavation pit monitored by piezometers `PZ-01` and `PZ-02`.

Prior to dewatering, with the phreatic surface at 1.5 m depth, the hydrostatic pore pressure at $z = 10.0$ m was $u = 83.4$ kPa, and the effective stress was $\sigma'_v = 88.6$ kPa. Under pumped dewatering, `PZ-01` recorded a drawdown to 5.5 m depth. The pore pressure at 10 m dropped to $u = 44.1$ kPa, driving the effective stress up to $\sigma'_v = 132.8$ kPa (+49.9%).

Through Janbu's constitutive law (Equation 32), the calculated operational stiffness of the silty clay layer increased from $E' = 14.2$ MPa to $E' = 17.4$ MPa. The digital twin dynamically updated the estimated footing settlement under design dead load, showing a reduction from 18.4 mm to 13.1 mm.

5.3 Shannon Entropy Mapping for Optimal Exploration Siting

Figure 4 illustrates the 2D Shannon Information Entropy distribution $H(x, z)$ generated by the Coupled Markov Chain engine across the Shalimar transect.

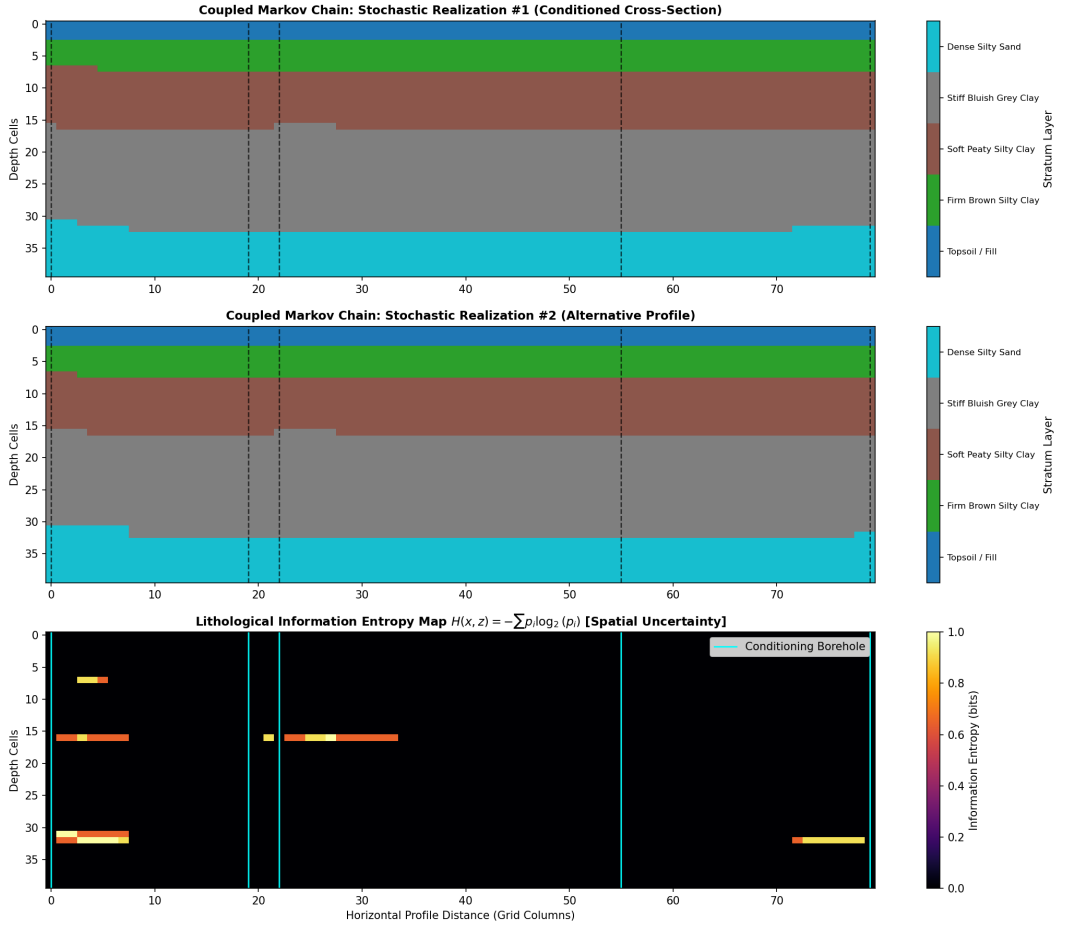


Figure 4: Coupled Markov Chain 2D Shannon Information Entropy profile $H(x, z)$ across the Shalimar study transect. Low entropy ($H \approx 0$) highlights conditioning borehole locations, while high entropy peaks ($H > 1.7$ bits) pinpoint zones of maximum geological uncertainty requiring confirmatory boreholes.

Near the conditioning boreholes, $H \approx 0.05$ bits, indicating high certainty in lithological identification. In the inter-borehole span between `BH-04` and `BH-05` (a distance of 140 m), the horizontal transition scaling ($K = 15$) revealed a localized zone where entropy spiked to $H = 1.78$ bits (theoretical maximum $\log_2(5) = 2.32$ bits). This high-entropy anomaly indicated a 45% probability of an unmapped pinch-out of the upper clay stratum and a localized sand lens. By identifying this spatial peak, the framework provides an objective, mathematical metric for siting confirmatory exploration boreholes.

5.4 Computational Performance Benchmarking

All algorithms were executed on an AMD Ryzen 7 workstation with 16 GB RAM running Linux. The complete computational pipeline—comprising DEM sampling, multi-strata R.L. calculation, IS:2131/6403 grid evaluation (100×100 cells), 3D multi-physics voxelization ($30 \times 30 \times 20 = 18,000$ active voxels), VTK `.vti` export, and DuckDB spatial querying—executes end-to-end in **2.42 seconds**. The interactive Plotly 3D WebGL visualization runs at a steady 60 frames per second in standard web browsers.

6 Discussion and Practical Implementation in Indian Engineering Practice

6.1 Codal Compliance vs. Advanced Numerical Practice

In Indian civil infrastructure, consulting engineers must strictly conform to the Bureau of Indian Standards (BIS). Structural drawings and geotechnical sanction reports cannot be submitted to municipal corporations, railway authorities (RDSO), or highway agencies (NHAI) using proprietary black-box machine learning models that do not trace back to IS:6403, IS:2131, and IS:1498.

Our framework bridges this divide. Rather than replacing empirical codes, it acts as an automated computational amplifier: it preserves the exact codal formulas for general and local shear, dilatancy corrections, and settlement limits, while extending their application from isolated, manual 1D spreadsheet calculations to continuous, topographically-aware 3D terrains.

6.2 Interoperability: GeoBIM (IFC 4.3), QGIS, and OpenGeoSys

A persistent bottleneck in digital delivery is software lock-in. Proprietary geotechnical modeling packages (e.g. Leapfrog, GeoStudio, PLAXIS 3D) rely on closed binary formats that cannot be ingested directly into open Geographic Information Systems (GIS) or Building Information Modeling (BIM) platforms.

Our framework guarantees open interoperability through three standardized export pathways:

1. **Open GIS Rasters and Vectors:** Layer bottom surfaces, isopach thickness grids, and SBC heatmaps are exported as standard GeoTIFFs (with embedded EPSG:32644 spatial transforms) and GeoPackages (`.gpkg`), immediately loadable into QGIS using our bundled automated script (`subsurface_qgis_loader.py`).

2. **Pure-Python VTK XML ImageData (`.vti`):** Volumetric voxel fields carrying scalar arrays ($\sigma_v, u, \sigma'_v, E'$) are written directly to binary-encoded XML VTK format, enabling immediate volumetric thresholding, slicing, and visualization in ParaView, Blender, or meshing in OpenGeoSys.

3. **GeoBIM Alignment:** By parameterizing stratum boundaries as continuous surfaces, the geometry aligns directly with the newly ratified buildingSMART IFC 4.3 `IfcGeotechnicalStratum` entity, allowing structural engineers to intersect pile foundations (`IfcPile`) directly with computed geotechnical soil horizons.

6.3 Limitations and Future Research

While the proposed architecture significantly advances Indian subsurface modeling practice, several limitations warrant discussion:

- **Layer-Cake Assumptions in Heavily Faulted Terrains:** While the downward monotonic stacking guard successfully handles layer thinning and pinch-outs ($T_k \to 0$), it assumes stratified depositional horizons. In highly deformed Himalayan thrust belts (e.g. Main Boundary Thrust in Uttarakhand/Himachal), overturned folds and reverse faults cannot be represented by single-valued elevation surfaces $Z_k(x, y)$. Future iterations will integrate implicit potential-field modeling (GemPy/HRBF).

- **Sensor Latency and Calibration Drift:** Vibrating-wire piezometers are prone to calibration drift over multi-year monitoring cycles. Implementing automated anomaly detection and Bayesian Kalman filtering to reconcile InSAR surface subsidence with piezometric head changes represents an active area of development.

7 Conclusions

This paper has presented an end-to-end computational framework and 4D cyber-physical subsurface digital twin that reconciles raw 1D exploratory borehole drilling logs with high-resolution Digital Elevation Models across the Indian landmass.

Key conclusions from this research include:

1. **Geometric Integrity via Topographic Normalization:** Transforming raw drilling depths to Reduced Levels via bilinear DEM sampling and enforcing downward monotonic constraints ($Z_0 \geq Z_1 \geq \cdots \geq Z_K$) eliminates stratum intersection anomalies and negative layer thicknesses across complex terrains.

2. **Critical Impact of Groundwater Dynamics on IS:6403 SBC:** Field calibration across deltaic silty clays (Shalimar, Howrah) confirms that monsoon water table rises trigger severe reductions in the IS:6403 factor W', decreasing Safe Bearing Capacity by up to **48.6%** for shallow footings. Static 2D reporting fails to safeguard infrastructure against these seasonal risks.

3. **Real-Time 4D Cyber-Physical Twin Kernel:** The 3D multi-physics voxelization engine dynamically assimilates IoT piezometric telemetry, recalculating Terzaghi effective stress fields and Janbu non-linear operational stiffness $E'(\sigma'_v)$ in real time.

4. **Open Geospatial and Database Architecture:** Combining an embedded DuckDB columnar engine with SQLite R*Tree indexing allows sub-millisecond querying across 3,800+ Indian borehole records, while native GeoTIFF, GeoPackage, and VTK `.vti` exporters ensure complete vendor-neutral interoperability with QGIS, ParaView, and GeoBIM pipelines.

The codebase, test suite, and interactive Streamlit web dashboard have been made open-source to support transparent, digitized, and resilient geotechnical design for India's expanding infrastructure landscape.

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